

Regular fair realizations exist, but require factorially many parameters

Proof with completed independent mathematical review

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Work in the reduced free nilpotent Lie algebra $\mathfrak{g}_n$ on $X_1, \dots, X_n$, where every bracket containing a repeated generator is zero. Its dimension is

$$d_n = \sum_{k=1}^n \binom{n}{k} (k-1)! \sim e(n-1)!.$$

A positive word $w = i_1 \cdots i_L$ has weighted endpoint map

$$\Phi_w(t) = \prod_{a=1}^L \exp(t_a X_{i_a}), \quad t_a > 0.$$

We call it regular at t when this map has differential rank d_n . This is a statement about varying face weights while keeping their order fixed, not about changing face labels inside that same order.

Lemma 1 (A short regular positive prefix). *There is a positive unit-weight word of length at most nd_n whose weighted endpoint map is regular at the all-one vector.*

Proof. In the reduced algebra $(\text{ad } X_i)^2 = 0$, so

$$\text{Ad}_{\exp X_i} = I + \text{ad } X_i.$$

Every nested bracket of length at most n is therefore an integer linear combination of vectors $\text{Ad}_v X_j$, where v is a positive word of length at most $n-1$. Since nested brackets span $\mathfrak{g}_n$, these vectors span the whole Lie algebra.

For a current positive prefix with endpoint P , let S be the span of its existing right-trivialized weight derivatives. A face labeled i following prefix Q has such derivative $\text{Ad}_Q X_i$. If $S \neq \mathfrak{g}_n$, the preceding spanning assertion, after conjugating by P , supplies a positive word v of length at most $n-1$ and a label j with $\text{Ad}_{Pv} X_j \notin S$. Append v and then one j . The newly appended j has that derivative, so the derivative span increases. Earlier derivatives are unchanged by appending a suffix. At most d_n steps, each adding at most n letters, give full rank. $\square$

Theorem 1 (Existence of a regular fair word). *For every n , some equal-composition permutation-fair word of total length $\exp(O(n \log(n+1)))$ is regular for its positive face-weight endpoint map. For that word, imposing the n fixed total-weight constraints leaves rank $d_n - n$ onto the fixed-composition signature manifold.*

Proof. Apply the established positive prefix-completion theorem in `order_doubling_prefix_completion.tex` to the word from Lemma 1. It extends literally to a finite permutation-fair equal-composition word. The original prefix derivatives still have full rank, so the completed

word is regular. Its common count is at most $(nd_n + 1) \exp(O(n \log(n + 1)))$, and multiplying by n retains total length $\exp(O(n \log(n + 1)))$.

The abelianization of the endpoint differential is precisely the vector of total-weight variations of the individual letters. For a Lie tangent vector with zero abelianization, any preimage under the surjective differential already has zero total-weight variation on each letter. Thus restriction to the fixed-count tangent space is surjective onto the $d_n - n$ dimensional commutator part. $\square$

Corollary 1 (A limitation of generic full-rank correction). *Every positive weighted word with full endpoint rank has at least $d_n = \exp(\Omega(n \log n))$ face-weight parameters, hence at least that many faces. The same face-count conclusion holds if its fixed-count differential is surjective onto the full fixed-count signature manifold. Consequently any hypothetical $\exp(O(n))$ -length fair construction must be singular for this endpoint map. A generic implicit-function argument that requires surjectivity onto the entire reduced Lie signature space cannot alone provide such an exponential construction.*

Proof. The unrestricted rank is at most the number L of face weights. With one fixed total per letter, the source dimension is $L - n$ and surjectivity requires $L - n \geq d_n - n$. Both give $L \geq d_n$. The dimension asymptotic follows by writing

$$\frac{d_n}{(n-1)!} = \sum_{j=0}^{n-1} \frac{n}{(n-j)j!}$$

and splitting the exponential-series tail, or by elementary dominated estimates on $j \leq n/2$ and the factorially small remaining terms. $\square$

Corollary 2 (Optimal asymptotic order among regular fair words). *Let L_n^{reg} be the minimum total length of a unit-weight permutation-fair word whose positive face-weight endpoint map has full rank. Then*

$$\log L_n^{\text{reg}} = \Theta(n \log n).$$

Proof. The dimension bound gives $L_n^{\text{reg}} \geq d_n$, and Theorem 1 supplies the matching logarithmic-order upper bound. $\square$

These results are compatible with singular fair realizations and their inherited singularity. They determine the asymptotic logarithmic order only for *regular* fair words; the unrestricted minimum may be smaller. The regular upper bound uses the order-doubling completion proved later in this research session.